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```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records.

```
print(df.head())
```

```
   Roll No   Name Department  Marks  Attendance
0      101  Aryan     BSc IT     99          98
1      102  Jasad     BSc CS     65          80
2      103   Kaif     BSc IT     88          92
3      104  Sopte     BSc CS     72          80
4      105  Dhruv     BSc IT     55          70
```

```
print(df.tail())
```

```
   Roll No   Name Department  Marks  Attendance
0      101  Aryan     BSc IT     99          98
1      102  Jasad     BSc CS     65          80
2      103   Kaif     BSc IT     88          92
3      104  Sopte     BSc CS     72          80
4      105  Dhruv     BSc IT     55          70
```

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information.

```
print(df.shape) #number of rows and columns
```

```
(5, 5)
```

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

```
print(df.dtypes)
```

```
Roll No        int64
Name          object
Department    object
Marks          int64
Attendance     int64
dtype: object
```

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000.

```
result = df[df["Attendance"]>=90]
print(result)
```

```
   Roll No   Name Department  Marks  Attendance
0      101  Aryan     BSc IT     99          98
2      103   Kaif     BSc IT     88          92
```

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price.

```
result = df.sort_values("Marks")
print(result)
```

```
   Roll No   Name Department  Marks  Attendance
4      105  Dhruv     BSc IT     55          70
1      102  Jasad     BSc CS     65          80
3      104  Sopte     BSc CS     72          80
2      103   Kaif     BSc IT     88          92
0      101  Aryan     BSc IT     99          98
```

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

```
print("Maximum Marks: ",df["Marks"].max())
print("Minimum Marks: ",df["Marks"].min())
print("Average Marks: ",df["Marks"].mean())
print("Total Marks: ",df["Marks"].sum())
```

```
Maximum Marks:  99
Minimum Marks:  55
Average Marks:  75.8
Total Marks:  379
```